

Problem 4.5 Solution: Missile Problem (EM Algorithm and Analysis for Poisson Mixture Model)

Part (a) EM Algorithm Derivation for Poisson Mixture Model

1. Model and Latent Variable Definition

Let the observed data $X = \{x_1, x_2, \dots, x_n\}$ (x_i is the number of missile hits in the i -th region) follow a **K-component Poisson mixture distribution**:

$$p(x_i = k \mid \theta) = \sum_{j=1}^K \pi_j \cdot \text{Poisson}(k \mid \lambda_j) = \sum_{j=1}^K \pi_j \cdot \frac{\lambda_j^k e^{-\lambda_j}}{k!}$$

Where:

- Parameter set $\theta = \{\pi_j, \lambda_j\}_{j=1}^K$, with $\pi_j \geq 0$, $\sum_{j=1}^K \pi_j = 1$ (mixture weights), and $\lambda_j > 0$ (Poisson rate parameter for the j -th component);
- Latent variables $Z = \{z_{ij}\}_{i=1, n; j=1, K}$, where $z_{ij} = 1$ indicates x_i comes from the j -th component, $z_{ij} = 0$ otherwise, and $\sum_{j=1}^K z_{ij} = 1$.

2. Complete Data Log-Likelihood

The complete data is (X, Z) , and its joint probability distribution is:

$$p(X, Z \mid \theta) = \prod_{i=1}^n \prod_{j=1}^K \left[\pi_j \cdot \frac{\lambda_j^{x_i} e^{-\lambda_j}}{x_i!} \right]^{z_{ij}}$$

Taking the natural logarithm (using $\log(ab) = \log a + \log b$, $\log(a^b) = b \log a$), we get the **complete data log-likelihood**:

$$\log p(X, Z \mid \theta) = \sum_{i=1}^n \sum_{j=1}^K z_{ij} [\log \pi_j - \lambda_j + x_i \log \lambda_j - \log(x_i!)]$$

Note: $\log(x_i!)$ is independent of parameters θ and can be ignored in optimization.

3. E-step: Compute Posterior Expectation of Latent Variables (Soft Assignment Weights)

The E-step aims to compute the **Q-function** (expected complete data log-likelihood given the current parameters):

$$Q(\theta; \theta^{\text{old}}) = \mathbb{E}_{Z|X, \theta^{\text{old}}} [\log p(X, Z | \theta)]$$

The key is to compute the posterior expectation of the latent variables $\gamma_{ij} = \mathbb{E}[z_{ij} | x_i, \theta^{\text{old}}]$ (i.e., the posterior probability that x_i belongs to the j -th component). Using Bayes' rule:

$$\gamma_{ij} = \frac{P(x_i | z_{ij}=1, \theta^{\text{old}}) P(z_{ij}=1 | \theta^{\text{old}})}{\sum_{k=1}^K P(x_i | z_{ik}=1, \theta^{\text{old}}) P(z_{ik}=1 | \theta^{\text{old}})}$$

Substituting the Poisson likelihood $P(x_i | z_{ij} = 1, \theta^{\text{old}}) = \frac{(\lambda_j^{\text{old}})^{x_i} e^{-\lambda_j^{\text{old}}}}{x_i!}$ and prior $P(z_{ij} = 1 | \theta^{\text{old}}) = \pi_j^{\text{old}}$, we simplify to:

$$\gamma_{ij} = \frac{\pi_j^{\text{old}} \cdot (\lambda_j^{\text{old}})^{x_i} e^{-\lambda_j^{\text{old}}}}{\sum_{k=1}^K \pi_k^{\text{old}} \cdot (\lambda_k^{\text{old}})^{x_i} e^{-\lambda_k^{\text{old}}}}$$

Substituting γ_{ij} into the Q-function, define:

- $N_j = \sum_{i=1}^n \gamma_{ij}$ (effective sample count for the j -th component);
- $S_j = \sum_{i=1}^n \gamma_{ij} x_i$ (weighted total hits for the j -th component);

Then the Q-function simplifies to:

$$Q(\theta; \theta^{\text{old}}) = \sum_{j=1}^K [N_j \log \pi_j - N_j \lambda_j + S_j \log \lambda_j]$$

4. M-step: Maximize Q-function to Update Parameters

The M-step updates π_j and λ_j by maximizing the Q-function.

(1) Update Mixture Weights π_j (Constrained Optimization)

Goal: Maximize $\sum_{j=1}^K N_j \log \pi_j$, subject to $\sum_{j=1}^K \pi_j = 1$.

Construct the Lagrangian:

$$\mathcal{L}(\pi, \lambda) = \sum_{j=1}^K N_j \log \pi_j + \lambda \left(1 - \sum_{j=1}^K \pi_j \right)$$

Take partial derivative w.r.t. π_j and set to zero:

$$\frac{\partial \mathcal{L}}{\partial \pi_j} = \frac{N_j}{\pi_j} - \lambda = 0 \implies \pi_j = \frac{N_j}{\lambda}$$

Substitute into the constraint $\sum_{j=1}^K \pi_j = 1$, yielding $\lambda = \sum_{j=1}^K N_j = n$ (since $\sum_{j=1}^K \gamma_{ij} = 1$, we have $\sum_{j=1}^K N_j = \sum_{i=1}^n \sum_{j=1}^K \gamma_{ij} = n$). Thus:

$$\pi_j^{\text{new}} = \frac{N_j}{n} = \frac{1}{n} \sum_{i=1}^n \gamma_{ij}$$

(2) Update Rate Parameters λ_j (Unconstrained Optimization)

Goal: Maximize $\sum_{j=1}^K (-N_j \lambda_j + S_j \log \lambda_j)$.

Take partial derivative w.r.t. λ_j and set to zero:

$$\frac{\partial Q}{\partial \lambda_j} = -N_j + \frac{S_j}{\lambda_j} = 0 \implies \lambda_j = \frac{S_j}{N_j}$$

Substituting definitions of S_j and N_j :

$$\lambda_j^{\text{new}} = \frac{\sum_{i=1}^n \gamma_{ij} x_i}{\sum_{i=1}^n \gamma_{ij}}$$

5. Connection to Single Poisson ML Estimate

Recall Problem 2.1: If all data comes from a **single Poisson distribution** ($K = 1$), then the posterior expectation $\gamma_{i1} = 1$ (each sample definitively belongs to the single component). Then:

- $N_1 = \sum_{i=1}^n 1 = n$, $S_1 = \sum_{i=1}^n x_i$;
- The ML estimate for the rate parameter is $\hat{\lambda} = \frac{S_1}{N_1} = \frac{1}{n} \sum_{i=1}^n x_i$, identical to the single Poisson ML estimate.

Thus, the M-step update λ_j^{new} is a **weighted version of the Poisson ML estimate**, with weights given by the soft assignment probabilities γ_{ij} from the E-step.

6. EM Algorithm Iteration Process

1. **Initialization:** Randomly set $\theta^0 = \{\pi_j^0, \lambda_j^0\}$ (satisfying $\sum \pi_j^0 = 1$, $\lambda_j^0 > 0$);
2. **E-step:** Compute all γ_{ij}^t using θ^t (formula in Section 3);
3. **M-step:** Update θ^{t+1} using γ_{ij}^t (formulas in Sections 4.1 and 4.2);
4. **Convergence Check:** If $\|\theta^{t+1} - \theta^t\| < \epsilon$ (or the observed data log-likelihood $\log p(X | \theta)$ converges), stop; else return to Step 2.

Part (b) Theoretical Analysis of Bombing Data (Based on Model Selection and Parameter Interpretation)

1. Data Preprocessing: Baseline Statistics

First, compute basic statistics for both cities (as benchmarks for single Poisson models):

City	Total Regions n	Total Hits $T = \sum_{k=0}^5 k \cdot N_k$	Single Poisson ML Estimate $\hat{\lambda}_1 = T/n$
London	576	535	$535/576 \approx 0.929$
Antwerp	576	516	$516/576 \approx 0.896$

2. Model Selection Criterion: BIC (Bayesian Information Criterion)

To determine whether a mixture model is better than a single Poisson model (i.e., whether target regions exist), use BIC to select the optimal K :

$$\text{BIC}(K) = -2 \log p(X \mid \hat{\theta}_K) + d_K \log n$$

Where:

- $\log p(X \mid \hat{\theta}_K)$: log-likelihood of observed data under optimal parameters $\hat{\theta}_K$ (goodness-of-fit);
- $d_K = 2K - 1$: number of model parameters (K λ_j 's + $K - 1$ independent π_j 's, since $\sum \pi_j = 1$);
- The optimal K minimizes BIC (balancing goodness-of-fit and model complexity).

3. Key Analysis: Physical Meaning of λ_j and π_j

In the Poisson mixture model, λ_j represents the "average hit count" for the j -th component, and π_j represents the "proportion of regions" belonging to that component. If **target regions** exist, the model must satisfy:

- At least one component has $\lambda_j \gg \hat{\lambda}_1$ (hit count significantly higher than average);
- That component's $\pi_j \in (0, 1)$ (reasonable proportion, not extremely small).

4. Theoretical Conclusion Derivation

(1) London

Assume BIC selects optimal $K = 2$ (significant improvement in goodness-of-fit, minimal BIC). Then parameters typically satisfy:

- $\lambda_1 \approx 0.7$ (non-target regions, hit count below single Poisson's 0.929), $\pi_1 \approx 0.8$ (80% non-target regions);
- $\lambda_2 \approx 2.0$ (target regions, hit count over twice the average), $\pi_2 \approx 0.2$ (20% target regions).

Reason: London data has $35 + 7 + 1 = 43$ regions with 3 or more hits (7.5%), requiring a high- λ component for fitting; and BIC shows $K = 2$ outperforms $K = 1$, indicating target regions exist.

(2) Antwerp

Assume BIC selects optimal $K = 1$ (BIC does not decrease significantly for $K \geq 2$). Then parameters satisfy:

- All $\lambda_j \approx 0.896$ (component hit counts close to single Poisson estimate), π_j show no clear separation (e.g., for $K = 2$, $\lambda_1 \approx 0.85$, $\lambda_2 \approx 0.95$, very small difference).

Reason: Antwerp data has 21 regions with 5 or more hits (3.6%), but can be fit by a single Poisson model (BIC does not support a mixture model), indicating no significant target regions.

5. Final Conclusion

1. **London:** Significant target regions exist (optimal $K = 2$, target region hit count over twice that of non-target);
2. **Antwerp:** No evidence of target regions (optimal $K = 1$, component hit counts show no significant difference).